Cisco Team 3 Time Management Document

Project Management & Tracking Approach

Cisco Team 3 will use Trello to track user stories, tasks, sprint progress, and acceptance criteria. GitHub will be used to organize project documentation, configurations, and supporting evidence. Progress will be reviewed during team meetings and reflected in applicable IT491 PM Reports.

Trello Link:
https://trello.com/invite/b/6abd4fa5848c7a14efff92be/ATTIe677aebed4b0511ddc7a58215ac4e6015EB5A6B2/it491-cisco-team-3-ccna-13

Github Link: https://github.com/benpopso/IT491-Team-3-Cisco-Team

Responsibility and Work Allocation

Work will be allocated per teammate based on their assigned roles. The team will break sprints into subtasks as they approach each sprint. An owner will be assigned to tasks who will be in charge of verifying completion ensuring the task is completed sufficiently. Contributors will be highlighted if other team members assisted in the process, and which tasks or to what extent they sasisted. This will help track individual contributions while priotizing teamwork to complete the project. Work assignments may be adjusted as project needs develop.

Sprint Plan

Note: This sprint plan establishes the initial implementation sequence based on the approved project scope, project dependencies, IT491 course milestones, and current MDDDE planning. Sprint end dates represent internal team planning and closeout targets and do not replace official IT491 submission deadlines. User stories, tasks, acceptance criteria, and individual work items may be refined, moved between sprints, or reprioritized as implementation, testing, sponsor feedback, and technical findings develop. Changes will be reflected in Trello and project documentation.

Additional Note: PM Reports will document project progress during each applicable reporting period. PM Report submission dates do not necessarily represent the completion date of an internal project sprint. Sprints will follow the planning dates established in this document, while Trello will reflect the current status of user stories, tasks, acceptance criteria, and supporting

evidence throughout each sprint. Official IT491 PM Reports and other course requirements will be submitted according to their designated IT491 course deadlines.

User stories can be found in Trello

Initial Sprint Plan

Sprint 1: Project Readiness, Design & Network Foundation
September 29 - October 12

Scope Week Alignment: Week 1 and Week 2

Goal: Complete applicable pre-project requirements, establish the project's architecture and standards, and build the segmented network foundation.

Planned Work

- Complete applicable IT491 pre-project and readiness requirements
- Complete and refine initial MDDDE planning document
- Confirm team roles and responsibilities
- Confirm team communication structure
- Establish Trello backlog and sprint structure
- Establish GitHub repository and document organization
- Establish initial project risks, dependencies, milestones, and project tracking
- Review project scope and establish initial requirements list
- Finalize initial logical topology
- Finalize initial physical topology
- Finalize addressing and segmentation plan
- Develop initial Build Standard
- Establish initial standards for naming, management access, hardening, and applicable services
- Conduct team review and agreement on architecture, addressing, segmentation, and Build Standard
- Create initial Acceptance Test Matrix
- Begin Packet Tracer topology
- Begin device naming and baseline configuration
- Complete VLAN configuration
- Complete trunk configuration
- Complete inter-VLAN routing

- Complete routed infrastructure and single-area OSPF as applicable
- Conduct capability spike confirming RADIUS, wireless, Syslog, and NTP behavior in this Packet Tracer version
- Complete and document required team meetings
- Complete applicable PM Report and readiness requirements

Initial User Story Placement

- US-01: Standardized Network Architecture
- US-02: Consistent Network Addressing
- US-03: VLANs
- US-04: VLAN Connectivity Across Switches
- US-05: Inter-VLAN Routing
- US-06: Routed Network Connectivity
- US-12: Verified Network Requirements
- US-16: Consistent Network Build Standard

Deliverables

- D1 Packet Tracer Simulation: Initial build
- D2 Network Topology & Addressing: Initial
- D3 Build Standard: Initial
- D4 Segmented & Routed Network: Completed
- Acceptance Test Matrix: Initial

Sprint 2: Centralized Services & Administrative Authentication

October 13 - October 26

Scope Week Alignment: Week 3 and Week 4

Goal: Deploy the centralized addressing and name-resolution services the network depends on, then layer in centralized administrative authentication and secure management access.

Planned Work

- Deploy centralized DHCP
- Verify appropriate addressing, gateway, and DNS information

- Deploy centralized DNS
- Verify internal name resolution
- Deploy NTP ahead of Syslog
- Verify network-device time synchronization
- Record implemented services against current Build Standard
- Deploy RADIUS
- Configure AAA for network-device administration
- Verify authorized administrative authentication through RADIUS
- Implement SSH for secure remote administration
- Verify required devices support secure management access
- Verify unencrypted remote management is refused
- Document the applicable local-authentication fallback and intended RADIUS approach if full integration cannot be completed
- Confirm management-access settings conform to the Build Standard
- Perform initial component-level testing
- Update Acceptance Test Matrix with Sprint 2 results
- Capture applicable supporting evidence
- Refine topology, addressing documentation, and Build Standard as implementation develops
- Complete and document required team meetings
- Complete applicable PM Report and course requirements

Initial User Story Placement

- US-09: Centralized DHCP Service
- US-10: Centralized DNS Service
- US-13: Centralized NTP Service
- US-24: Automatic Network Configuration
- US-25: Internal Resource Name Resolution
- US-07: Central Admin Authentication
- US-08: Secure Remote Administration
- US-16: Consistent Network Build Standard, refinement

Deliverables

- D1 Packet Tracer Simulation: Functioning network foundation
- D2 Network Topology & Addressing: Refined as necessary
- D3 Build Standard: Refined
- D5 Centralized Services: DHCP, DNS, and NTP functioning
- D6 Authentication & Secure Administration: Initial completed implementation
- Acceptance Test Matrix: Updated with applicable service and authentication results

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Sprint 3: Access Control, External Connectivity, Monitoring, Wireless & Midterm MVP
October 27 - November 9

IT491 Midterm Submission: November 8

Scope Week Alignment: Week 5 and Week 6

Goal: Enforce inter-segment access control, establish outbound external connectivity, deploy centralized monitoring and secure wireless, complete access-layer hardening, and produce a working MVP for the IT491 Midterm.

Planned Work

- Implement extended ACLs according to the established segmentation and access requirements
- Verify applicable access relationships in permitted and denied directions
- Verify ACLs do not unintentionally block required DHCP, DNS, or RADIUS traffic
- Implement NAT/PAT for outbound external connectivity
- Verify simulated outbound external connectivity
- Perform applicable cross-training between team members
- Deploy centralized Syslog
- Verify centralized logging receives applicable network-device events
- Verify Syslog uses NTP-synchronized timestamps
- Implement secure wireless connectivity
- Integrate required RADIUS-backed wireless authentication
- Integrate wireless clients with applicable DHCP service
- Apply applicable ACL policy to wireless connectivity
- Verify permitted and restricted wireless connectivity
- Configure Port Security on applicable access ports
- Configure PortFast where appropriate
- Configure BPDU Guard where appropriate
- Secure unused access ports
- Confirm hardening settings conform to the Build Standard
- Continue testing implemented components
- Update Acceptance Test Matrix with Sprint 3 results
- Capture applicable evidence
- Refine Build Standard where implementation requires it

-Update topology and addressing documentation if necessary

- Review current project functionality against applicable Midterm requirements
- Organize current Trello, GitHub, MDDDE, testing, and technical evidence for Midterm
- Complete and document required team meetings
- Complete applicable PM Report and Midterm requirements

Initial User Story Placement

- US-18: Network Access Control
- US-19: ACL and Network-Service Integration
- US-26: External Resource Access
- US-14: Centralized Syslog Service
- US-20: Secure Wireless Access
- US-27: Wireless Employee Connectivity
- US-28: Time-Synchronized Syslog
- US-21: Port Security
- US-22: Unused-Port Protection
- US-23: Access-Layer Switching Protection

Continued throughout Sprint 3:

- US-12: Verified Network Requirements
- US-16: Consistent Network Build Standard

Midterm MVP Target

By the end of Sprint 3, the target MVP includes:

- Working Packet Tracer enterprise topology
- VLANs and trunking
- Inter-VLAN routing
- Applicable single-area OSPF and routed connectivity
- DHCP
- DNS
- NTP
- RADIUS/AAA
- SSH administration
- Extended ACLs
- NAT/PAT
- Centralized Syslog with NTP-correct timestamps
- Secure wireless connectivity
- Access-layer baseline and hardening implemented to the current standard
- Initial permit and deny testing

- Current logical and physical topology
- Current addressing documentation
- Current Build Standard
- Acceptance Test Matrix populated for functionality implemented through the Midterm
- Current Trello, GitHub, MDDDE, and project evidence

Deliverables

- D1 Packet Tracer Simulation: Midterm MVP
- D2 Network Topology & Addressing: Current
- D3 Build Standard: Current
- D4 Segmented & Routed Network: Completed
- D5 Centralized Services: Completed
- D6 Authentication & Secure Administration: Verified
- D7 Access Control, Wireless & Hardening: Completed
- D8 Integration & Interoperation Evidence: Initial evidence as applicable
- Acceptance Test Matrix: Updated through Midterm functionality

Milestone: IT491 Midterm MVP, November 8

The current working MVP and applicable project-management, MDDDE, documentation, testing, and technical evidence will be prepared for the IT491 Midterm submission.

Sprint 4: Integration, Induced-Failure Testing, Non-Author Audit & Final Delivery

November 10 - November 23

Project Completion Target: November 24

Scope Week Alignment: Week 7 and Week 8

Goal: Prove the integrated environment works as one coherent whole under cross-service and induced-failure testing, complete the non-author audit, then freeze, finalize, and package the project for the November 24 completion target.

Planned Work

- Execute the cross-service end-to-end journey
- Verify required services and security controls operate together
- Verify applicable permitted and denied connectivity
- Conduct ACL induced-failure testing
- Observe and diagnose the resulting service disruption
- Repair and retest the ACL dependency
- Conduct NTP induced-failure testing
- Observe and diagnose the resulting Syslog timestamp inconsistency
- Restore NTP and verify timestamp alignment
- Maintain integration-conflict log
- Document significant integration problems and their resolutions
- Conduct non-author audit against current Build Standard
- Record pass/fail results for audited components
- Correct applicable nonconformities and integration defects
- Perform integration and regression testing
- Update Acceptance Test Matrix
- Capture cross-service, induced-failure, audit, and integration evidence
- Freeze the technical configuration once final corrections are completed
- Perform final regression testing
- Complete the full Acceptance Test Matrix
- Verify applicable acceptance tests have expected and actual result**s**
- Verify required evidence is organized and traceable
- Perform final cross-service verification
- Perform final permit and deny verification
- Verify centralized services remain functional with full ACL policy applied
- Verify final routing, ACL, and Build Standard consistency
- Resolve or document remaining defects, nonconformities, or known limitations
- Finalize integration-conflict log
- Finalize non-author audit results
- Finalize logical and physical topology diagrams
- Finalize addressing documentation
- Finalize Build Standard
- Finalize troubleshooting documentation
- Finalize Packet Tracer file
- Verify required device configurations are saved appropriately
- Export required device configurations
- Organize final evidence and repository contents
- Verify contribution evidence from all team members
- Complete and document required team meetings
- Complete applicable PM Report and course requirements

Initial User Story Placement

- US-11: Integrated Network Operation
- US-15: Network Audit
- US-29: ACL Dependency Troubleshooting
- US-30: NTP Failure Troubleshooting
- US-32: Integration Conflict Documentation
- US-33: Protection of Sensitive Business Information
- US-34: Reliable Business-Supporting Network Operation
- US-17: Verifiable Project Completion
- US-31: Network Troubleshooting Documentation
- US-35: Verifiable Network Protection

Continued verification/refinement:

- US-12: Verified Network Requirements
- US-16: Consistent Network Build Standard
- US-18: Network Access Control
- US-19: ACL and Network-Service Integration
- US-21: Port Security
- US-22: Unused-Port Protection
- US-23: Access-Layer Switching Protection

Deliverables

- D1 Packet Tracer Simulation: Complete
- D2 Network Topology & Addressing: Complete
- D3 Build Standard: Complete
- D4 Segmented & Routed Network: Complete and verified
- D5 Centralized Services: Complete and verified
- D6 Authentication & Secure Administration: Complete and verified
- D7 Access Control, Wireless & Hardening: Complete and verified
- D8 Integration & Interoperation Evidence: Complete
- D9 Non-Author Audit: Complete
- Acceptance Test Matrix: Complete
- D10 technical materials and evidence: Available for final packaging

November 24 Project Completion Target

By November 24, the team's target is for the technical network and associated project evidence to be complete, as reflected in the Deliverables above.

Sprint 5: Final Delivery & Showcase Preparation

November 24 - December 3

Final Showcase: December 5

Scope Week Alignment: Following Week 8, IT491 final Showcase preparation

Goal: With the technical network and evidence already complete as of the November 24 target, finalize the report and presentation materials, record the video demonstration, and rehearse for Showcase.

Planned Work

- Finalize consolidated project report
- Finalize presentation slide deck
- Prepare final technical demonstration
- Record required project video
- Demonstrate the enterprise environment and required network segments in the video
- Demonstrate the cross-service end-to-end journey in the video
- Demonstrate required induced-failure behavior and repair
- Include applicable NTP/Syslog failure testing
- Include applicable non-author audit evidence
- Ensure all four team members narrate applicable portions of the video
- Ensure all four team members can explain the complete environment
- Rehearse final presentation
- Conduct team question-and-answer preparation
- Organize final submission package
- Perform final repository and file checks
- Make minor corrections discovered during final review
- Prepare for the December 5 Showcase

Final verification and evidence as applicable:

- US-01: Standardized Network Architecture
- US-02: Consistent Network Addressing

- US-11: Integrated Network Operation
- US-12: Verified Network Requirements
- US-15: Network Audit
- US-16: Consistent Network Build Standard
- US-28: Time-Synchronized Syslog
- US-32: Integration Conflict Documentation
- US-33: Protection of Sensitive Business Information
- US-34: Reliable Business-Supporting Network Operation

Earlier completed stories may also receive final regression testing or evidence updates without reopening their implementation unless a defect is identified.

Final Deliverables

- D1 Packet Tracer Simulation: Final
- D2 Network Topology & Addressing: Final
- D3 Build Standard: Final
- D4 Segmented & Routed Network: Final
- D5 Centralized Services: Final
- D6 Authentication & Secure Administration: Final
- D7 Access Control, Wireless & Hardening: Final
- D8 Integration & Interoperation Evidence: Final
- D9 Non-Author Audit: Final
- D10 Exported Configurations, Final Report, Presentation & Video Demonstration: Final
- Acceptance Test Matrix: Final